

SHI FENG

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EDUCATION

Tsinghua University

Sep 2019 - Jul 2023(expected)

Bachelor of Science, selected to Yao Class

- GPA: 3.87/4.0
- GRE Scores: Verbal Reasoning 157, Quantitative Reasoning 170, Analytical Writing 4.0
- Selected Courses: Introduction to AI (A), Mathematics for Computer Science (A), Causal and Statistical Learning (A+), Abstract Algebra (A), Introduction to Databases (A+), Artificial Intelligence: Principles and Techniques (A), Theory of Computation (A), Distributed and Blockchain System (A-), Machine Learning (A), Operating System (A)

SELECTED HONORS & AWARDS

“Star of Tomorrow” Award of Excellence

Mar 2022, MSR Asia

Best Paper Award(1/57)

Nov 2021, CSoNet

Innovation Excellence Award(top 10%)

Oct 2021, THU

“Challenge Cup” Science and Technology Competition, Top Grade Prize(top 3%)

Apr 2021, THU

Comprehensive Excellence Award(top 10%)

Oct 2020, THU

Russian Mathematical Olympiad, Prizewinner

Apr 2019, MSE of Russia

Chinese Mathematical Olympiad, Gold Medal

Nov 2018, CMS

National High School Mathematics League, First Prize

Sep 2016, 2017, 2018, CMS

PUBLICATION

Shi Feng, Wei Chen. **Causal Inference for Influence Propagation—Identifiability of the Independent Cascade Model**. *International Conference on Computational Data and Social Networks (CSoNet) 2021*. [[arXiv](#)] [[DOI](#)]

Shi Feng, Wei Chen. **Combinatorial Causal Bandits**. In submission. [[arXiv](#)]

Shi Feng, Fang-Yi Yu, Yiling Chen. **Peer Prediction for Learning Agents**. In submission. [[code](#)]

RESEARCH INTERESTS

Theoretical computer science and economics, especially in machine learning, algorithmic game theory, causality and learning theory.

RESEARCH EXPERIENCES

Peer Prediction for Learning Agents

Feb 2022 - Jul 2022

Advisors: Prof. [Yiling Chen](#), Prof. [Fang-Yi Yu](#)

Harvard EconCS Group

- Surveyed truthful mechanisms of information elicitation with consistent strategy assumption on agents.
- Proved truthful convergence of correlated agreement (CA) mechanism adopted on agents using reward-based online learning algorithms.
- Conducted numerical experiments which supported theoretical results.

Combinatorial Causal Bandits

Jul 2021 - Feb 2022

Advisor: Prof. [Wei Chen](#)

MSR Asia Theory Group

- Surveyed algorithms of causal bandits for both pure exploration and cumulative regret minimization.
- Proposed the new combinatorial causal bandit framework with its definition of cumulative regret.
- Proposed BGLM-OFU algorithm for combinatorial action space on Markovian binary generalized linear models.
- Proposed BLM-LR algorithm for combinatorial action space on binary linear models with unobserved nodes.

Causal Inference for Social Networks

Nov 2020 - Jul 2021

Advisor: Prof. Wei Chen

MSR Asia Theory Group

- Surveyed identifiability of Bayesian causal models and completeness of do-calculus.
- Provided transformations from an Independent Cascade Model to a Bayesian causal model.
- Proved necessary and sufficient conditions for identifiability in three common types of Independent Cascade Models.

ACADEMIC ACTIVITIES & SERVICES

Reviewer, Journal of Combinatorial Optimization (JOCO)

May 2022

Presenter, International Conference on Computational Data and Social Networks (CSoNet) ([slides](#))

Nov 2021

SKILLS

Programming Languages: C/C++, Python, Golang, SQL, Assembly language, Bash

Professional Applications: Latex, Mathematica, Matlab, PyTorch, TensorFlow, Git

Languages: Chinese (native language), English (fluent, 106 in TOEFL iBT)